

Advanced Environmental Modelling

Distributed non-linear lag models, spatially varying coefficients, and spatial DLNMs

Garyfallos Konstantinoudis

g.konstantinoudis@imperial.ac.uk

Imperial College London

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Advanced Environmental Models

- Distributed non-linear lag models
- Spatially varying coefficients
- Spatial DLNMs

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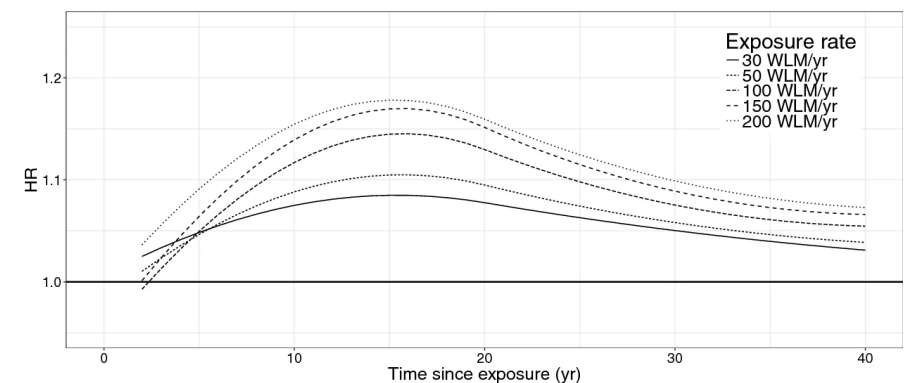
Distributed non-linear lag models

- An exposure event is frequently associated with a risk lasting for a defined period in the future.
- The risk at a given time is assumed a result of protracted exposures experienced in the past.
- Examples include, drugs, carcinogens, etc.

Challenge: The risk should be modelled in terms of contributions depending on intensity and timing of the exposure events. - Can use bi-dimensional association (interaction).

3

Example 1: Lung cancer and radon exposure



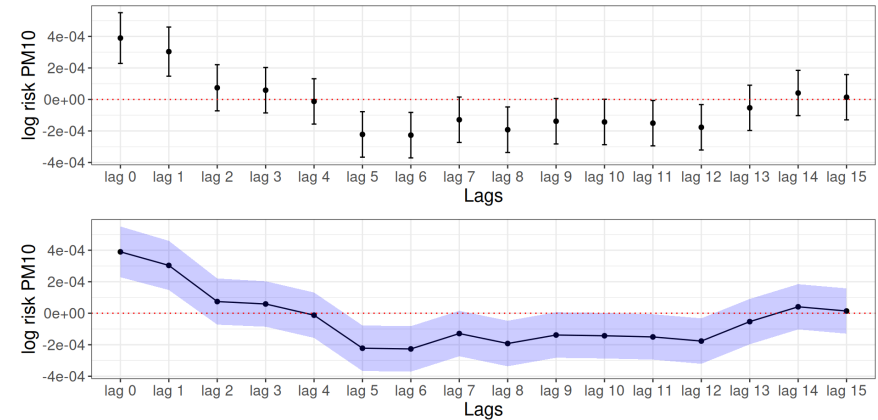
4

Example 2: PM10 in Chicago

	date	time	year	month	doy	dow	death	cvd	resp	temp	dptp
1	1987-01-01		1987	1	1	Thursday	130	65	13	-0.2777778	31.500
2	1987-01-02		1987	1	2	Friday	150	73	14	0.5555556	29.875
3	1987-01-03		1987	1	3	Saturday	101	43	11	0.5555556	27.375
4	1987-01-04		1987	1	4	Sunday	135	72	7	-1.6666667	28.625
5	1987-01-05		1987	1	5	Monday	126	64	12	0.0000000	28.875
6	1987-01-06		1987	1	6	Tuesday	130	63	12	4.4444444	35.125
	rhum	pm10	o3	pm10_laggeg							
1	95.500	26.95607	4.376079	NA							
2	88.250	NA	4.929803	NA							
3	89.500	32.83869	3.751079	NA							
4	84.500	39.95607	4.292746	NA							
5	74.500	NA	4.751079	NA							
6	77.375	40.95607	6.334412	NA							

Example 2: PM10 in Chicago

What are the main assumptions here? (Spoiler: independence!!)



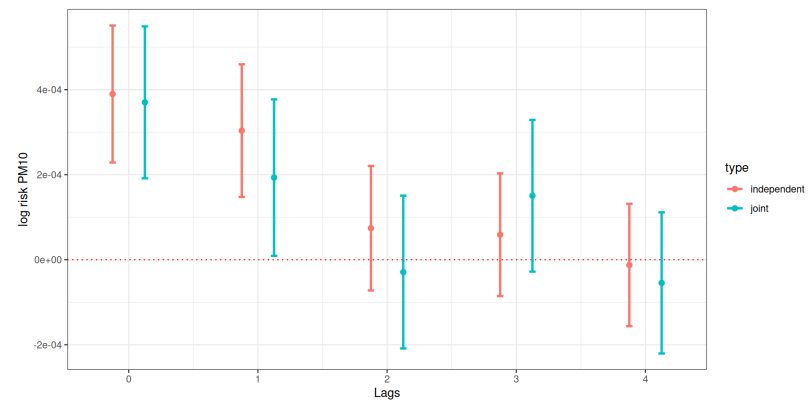
Distributed linear lag models

The unconstrained distributed lag model of order q is:

$$Y_t = \beta_0 + \beta_{10}X_t + \beta_{11}X_{t-1} + \dots + \beta_{1q}X_{t-q} + \epsilon_t$$

- $\beta_{1\ell}$ is the effect at lag $\ell = 0, 1, \dots, q$ and ϵ_t an error term.
- The overall impact for a unit change in X is given by $\sum_{\ell=0}^q \beta_{1\ell}$.

Example 2: PM10 in Chicago



Considerations

- Easy implementation when lags are few; overparametrized when we want to assess a lot of lags.
- Collinearity issues: The exposure is likely to be highly correlated with the values of the previous/after days. Weird behaviours in the point estimates (surprising protective effects), variance inflation.

Alternative: impose some constraints!

- A constant effect within lag intervals.
- Average of the exposures in the previous L day.
- Describing the coefficients with a smooth curve using continuous functions such as splines, polynomials, and other basis functions.

The idea: β_ℓ can be modelled using a basis function.

Polynomial DLM

Let $\beta_\ell = \sum_j \tau_j \ell^j$, $\ell = 0, \dots, q$, let's write it for 2 lags using a 3rd degree polynomial to see it explicitly:

$$Y_t = \beta_0 + \beta_{10}X_t + \beta_{11}X_{t-1} + \beta_{12}X_{t-2} + e_t$$

$$\beta_{10} = \tau_0, \beta_{11} = \tau_0 + \tau_1 + \tau_2 + \tau_3, \beta_{12} = \tau_0 + \tau_1 2 + \tau_2 2^2 + \tau_3 2^3$$

and we can modify as per first lecture to model more localized structures using: $\beta_\ell = \sum_j \tau_j \ell^j + \sum_k v_k (\ell - \kappa_k)_+^3$, thus:

$$\beta_{10} = \tau_0 + v_1(0 - \kappa_1)_+^3 + \dots + v_K(0 - \kappa_K)_+^3,$$

$$\beta_{11} = \tau_0 + \tau_1 + \tau_2 + \tau_3 + v_1(1 - \kappa_1)_+^3 + \dots + v_K(1 - \kappa_K)_+^3,$$

$$\beta_{12} = \tau_0 + \tau_1 2 + \tau_2 2^2 + \tau_3 2^3 + v_1(2 - \kappa_1)_+^3 + \dots + v_K(2 - \kappa_K)_+^3$$

and similarly we can penalize it can estimate *the penalised spline distributed lag estimate* of β_ℓ

Polynomial DLM in R: Chicago

```
1 cb1.pm <- crossbasis(chicagoNMMAPS$pm10,
2   lag = 15, argvar = list(fun = "lin"),
3   arglag = list(fun = "poly", degree = 4)
4 )
5 head(cb1.pm)
```

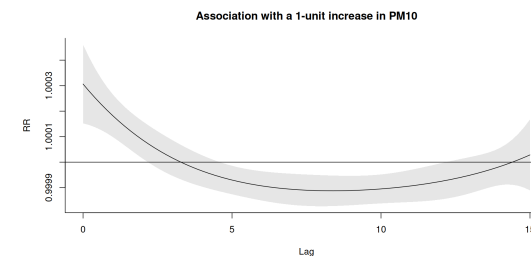
	v1.l1	v1.l2	v1.l3	v1.l4	v1.l5
[1,]	NA	NA	NA	NA	NA
[2,]	NA	NA	NA	NA	NA
[3,]	NA	NA	NA	NA	NA
[4,]	NA	NA	NA	NA	NA
[5,]	NA	NA	NA	NA	NA
[6,]	NA	NA	NA	NA	NA

```
1 cb1.pm[100:106,]
```

	v1.l1	v1.l2	v1.l3	v1.l4	v1.l5
[1,]	491.4144	234.4814	171.9688	138.7484	117.62724
[2,]	472.4144	228.8892	164.5070	132.7768	112.95375
[3,]	462.4144	217.7637	151.6641	120.3782	101.30370
[4,]	470.7431	214.8555	146.7695	115.3677	96.77336
[5,]	466.7431	214.2852	143.4258	111.3702	92.93659
[6,]	476.1822	224.1150	151.3660	118.8315	100.88929
[7,]	459.4839	217.5073	142.4543	108.5967	90.34264

Polynomial DLM in R: Chicago

```
1 pred1.pm <- crosspred(cb1.pm, model_dlm, at = 0:20, bylag = 0.2)
2 plot(pred1.pm,
3   ptype = "slices", var = 1, cumul = FALSE, ylab = "RR",
4   main = "Association with a 1-unit increase in PM10"
5 )
```



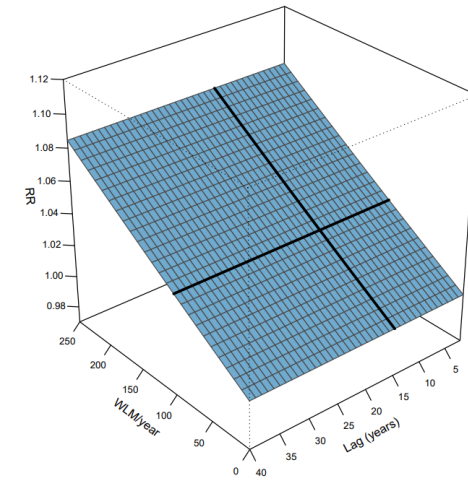
What is the main assumption here? Can we relax it?

Extension to distributed non-linear lag models

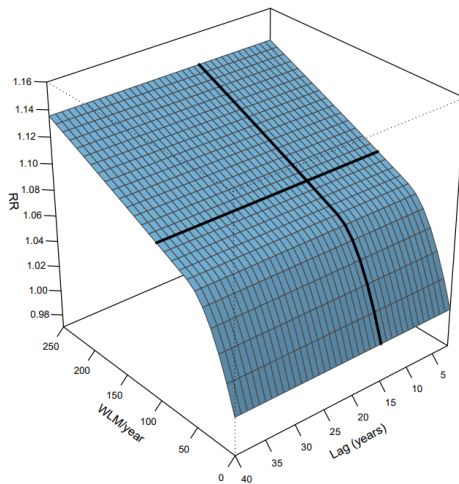
- We know that temperature and mortality have a U-shape relationship.
- We know that high temperature has a lag effect on mortality.
- Can we define models to combine these two components?

The idea: To calculate this bi-dimensional relationship, we need a basis function that combines the basis function in the lag dimension and the basis function in the exposure dimension. - We can use a cross-basis function.

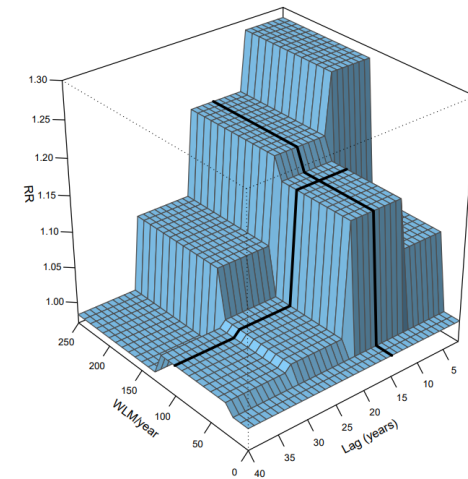
Linear-by-constant



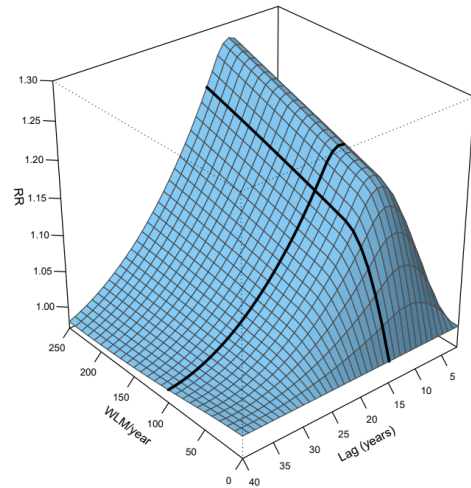
Spline-by-constant



Step-by-step



Spline-by-spline



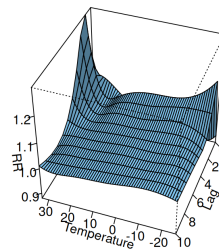
Example 3: Temperature in Chicago

```
1 cb2.pm <- crossbasis(chicagoNMMAPS$pm10,
2   lag = 1, argvar = list(fun = "lin"),
3   arglag = list(fun = "strata")
4 )
5
6 varknots <- equalknots(chicagoNMMAPS$temp, fun = "bs", df = 5, degree = 2)
7 lagknots <- logknots(10, 3)
8 cb2.temp <- crossbasis(chicagoNMMAPS$temp, lag = 10, argvar = list(
9   fun = "bs",
10  knots = varknots
11 ), arglag = list(knots = lagknots))
12
13
14 model_dlm2 <- mgcv::gam(death ~ cb2.pm + cb2.temp + s(time) + s(month) + dc
15   family = poisson(), chicagoNMMAPS
16 )
17
18 pred2.temp <- crosspred(cb2.temp, model_dlm2, cen = 21, by = 1)
```

Example 3: Temperature in Chicago

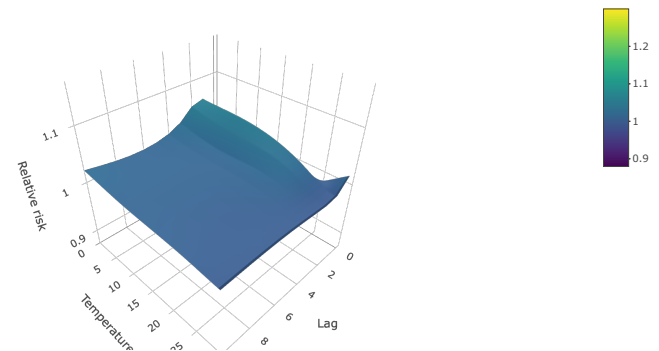
```
1 plot(pred2.temp,
2   xlab = "Temperature", zlab = "RR", theta = 200, phi = 40, lphi = 100,
3   main = "3D graph of temperature effect"
4 )
```

3D graph of temperature effect



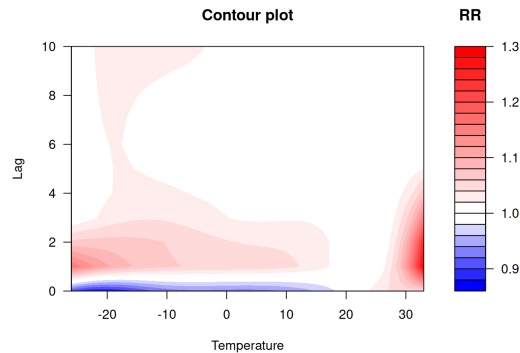
Example 3: Temperature in Chicago

```
1 library(plotly)
2 p <- plot_ly()
3 p <- add_surface(p, x = 0:10, y = -20:30, z = pred2.temp$matRRfit)
4
5 layout(p, margin = list(l = 0, r = 0), scene = list(
6   xaxis = list(title = "Lag"),
7   yaxis = list(title = "Temperature", range = c(0, 30)),
8   zaxis = list(title = "Relative risk")
9 ))
```



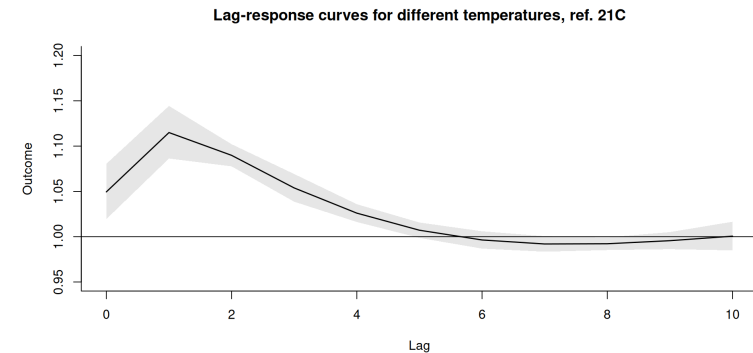
Example 3: Temperature in Chicago

```
1 plot(pred2.temp, "contour",
2     xlab = "Temperature", key.title = title("RR"),
3     plot.title = title("Contour plot", xlab = "Temperature", ylab = "Lag")
4 )
```



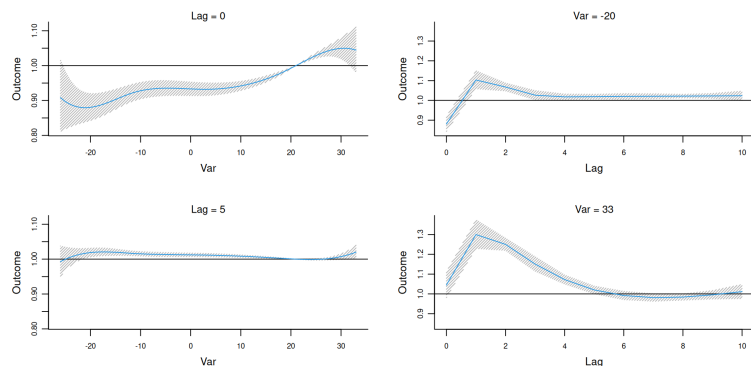
Example 3: Temperature in Chicago

```
1 plot(pred2.temp, "slices",
2     var = 30, col = 1, ylim = c(0.95, 1.2), lwd = 1.5,
3     main = "Lag-response curves for different temperatures, ref. 21C"
4 )
```



Example 3: Temperature in Chicago

```
1 plot(pred2.temp, "slices",
2     var = c(-20, 33), lag = c(0, 5), col = 4,
3     ci.arg = list(density = 40, col = grey(0.7))
4 )
```

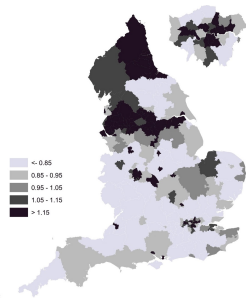


Spatially varying coefficients

- So far, we assumed a single effect of an exposure across the study region.
- However, different populations might experience different vulnerabilities to the effect of the exposure.
- If these vulnerabilities have spatial structures, then the effect of the exposure will vary in space.
- Spatial effect modification or spatial interaction.

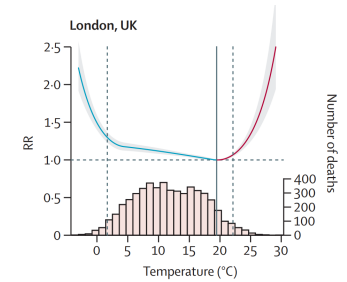
Example 1: Temperature and COPD

- 3rd cause of death, 3.17 million deaths in 2015 globally,
- In England, 115,000 emergency admissions and 24,000 deaths per year.
- COPD exacerbations: Bacteria, viruses and air-pollution.
- The role of temperature is unclear.



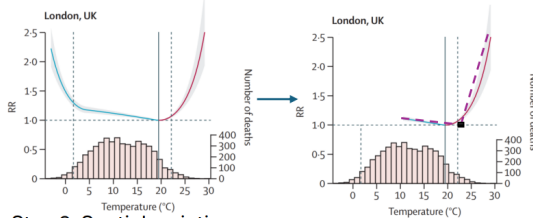
Example 1: Temperature and COPD

- Typically U-shaped relationship between temperature and health.
- Individual effect modifiers: age, sex, chronic conditions.
- Temporal variation of the effect (adaptation).
- Environmental effect modifiers: green space, urban heat island.
- Spatial vulnerabilities to heat exposure.

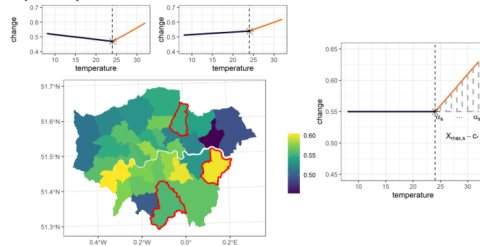


Example 1: Temperature and COPD

Step 1: Simplify the relationship



Step 2: Spatial variation

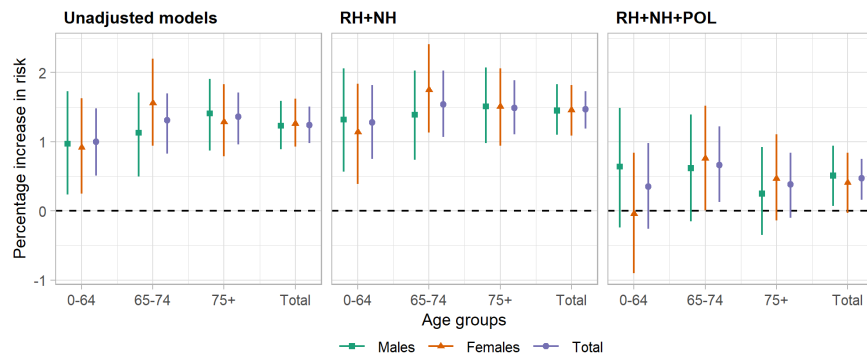


Example 1: Temperature and COPD

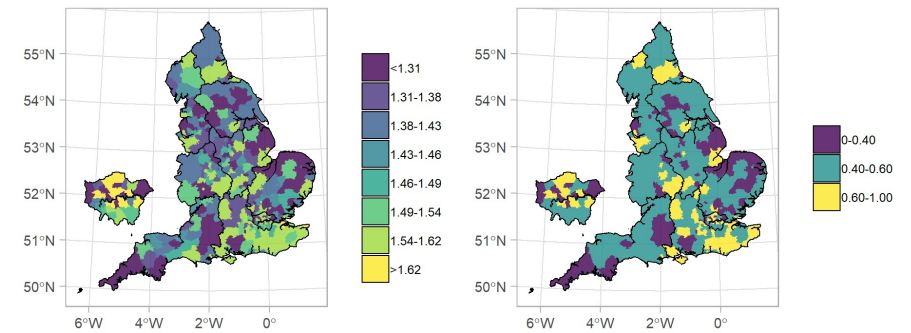
$$\begin{aligned}
 Y_{ijk} &\sim \text{Poisson}(\mu_{ijk}) \\
 \log(\mu_{ijk}) &= \alpha_1 I(X_{1i} < c) X_{1i} + \\
 &\quad \alpha_{2s} I(X_{1i} \geq c) X_{1i} + \sum \beta_m Z_{mi} + u_j + w_k \\
 \alpha_{2s} &= \alpha_2 + \sum \gamma_l H_l + b_s
 \end{aligned}$$

- Where should we put priors?
- How should we select the priors?

Example 1: Temperature and COPD



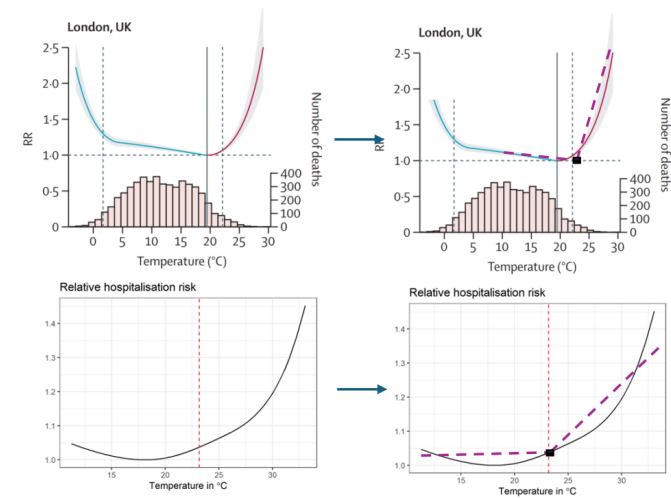
Example 1: Temperature and COPD



Example 1: Temperature and COPD

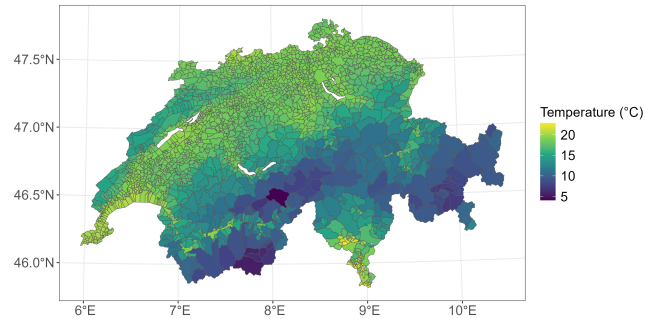
Effect modifier	Percentage increase	Pr
Green space	-1.46 (-6.99, 4.39)	0.30
Average temperature	-0.41 (-1.49, 0.71)	0.22
IMD Q1	1	-
IMD Q2	0.81 (-1.16, 3.08)	0.78
IMD Q3	1.57 (-0.76, 4.06)	0.91
IMD Q4	0.75 (-1.68, 3.36)	0.71
IMD Q5	1.62 (-1.31, 4.49)	0.85
Predominantly Rural	1	-
Urban with significant rural	-0.79 (-3.10, 1.51)	0.25
Predominantly urban	-1.57 (-4.16, 0.96)	0.12

Example 1: Temperature and COPD



Modelling the spatially varying non-linear effects of heat exposure

Main issues: (1) data sparsity, (2) spatial correlation



The model

We can write:

$$O_i \sim \text{Poisson}(\lambda_i E_i)$$

$$\log \lambda_i = \alpha + \sum_j (\beta_{1j} + \delta_{ij}) X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \theta_i + \phi_i$$

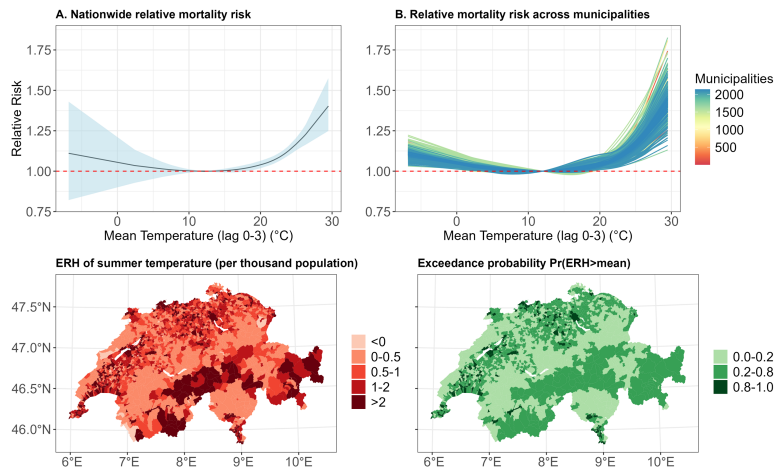
$$\theta_i \sim \text{Normal}(0, \sigma_\theta^2)$$

$$\phi \sim \text{ICAR}(W, \sigma_\phi^2)$$

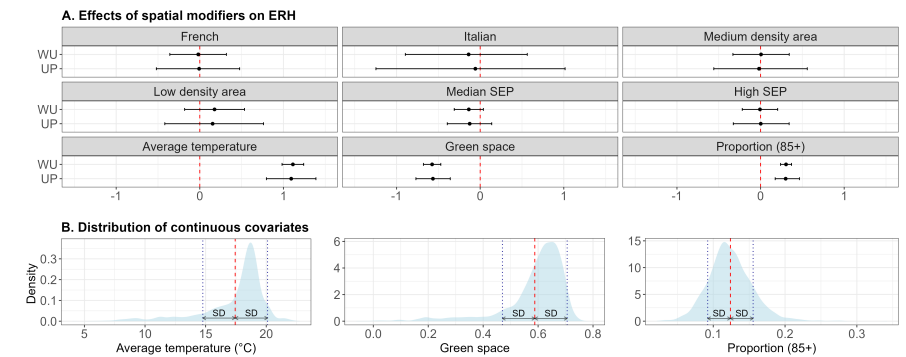
$$\delta_j \sim \text{ICAR}(W, \sigma_{\delta_j}^2)$$

And a step further: $\text{heat-risk}_i = \gamma_0 + \sum \gamma_i Z_i + w_i$

Beyond linear models: Case study



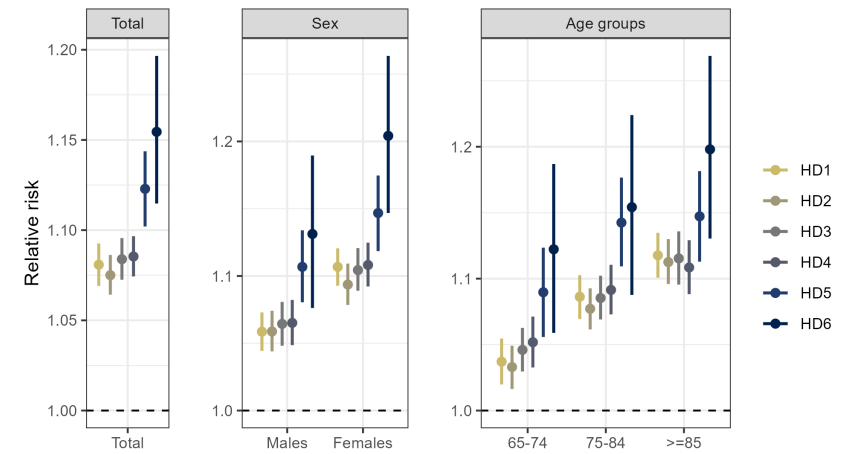
Beyond linear models: Effect modifiers



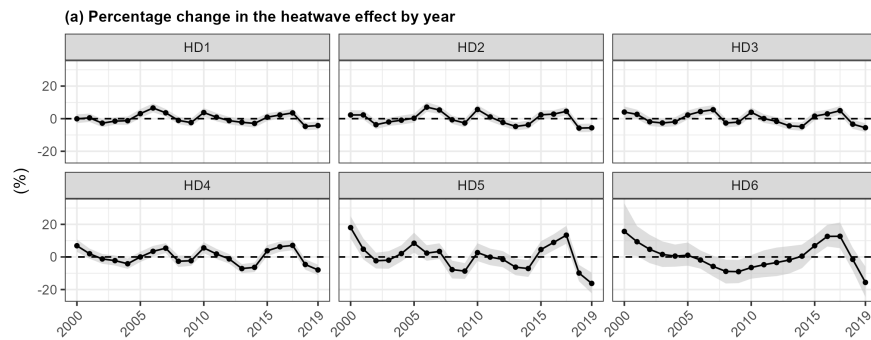
Spacetime-varying coefficients: Heatwaves in Greece

$$\begin{aligned}
 O_i &\sim \text{Poisson}(\lambda_i E_i) \\
 \log \lambda_{it} &= \alpha + (\beta_1 + \delta_i + u_i) X_{1it} + \\
 &\quad \beta_2 X_{2it} + \dots + \beta_k X_{kit} + \theta_i + \phi_i + v_i \\
 \theta_i &\sim \text{Normal}(0, \sigma_\theta^2) \\
 \phi, \delta &\sim \text{ICAR}(W, \sigma_\phi^2) \\
 u, v &\sim \text{RW2}(\sigma_u^2)
 \end{aligned}$$

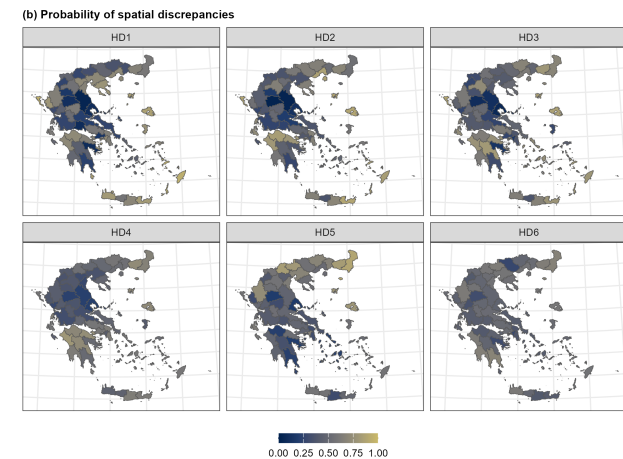
Overall results



Results in space



Results in time



Summary

- Introduction to distributed linear (and non-linear) lag models
- Introduction to spatially varying coefficients
- Spatial DLNMs
- Examples using air-pollution, radon and temperature

Getting ready for the lab

The lab for this session

- This lab will involve taking the spatiotemporal and non-linear models from the Advanced Environmental Modelling lecture and fitting them in [INLA](#), which is what makes models of this size practical to run at all.
- During this lab session, we will:
 1. Explore dengue case data for 645 Brazilian micro-regions from 2009 to 2024, alongside temperature, rainfall and humidity from ERA5;
 2. Build the neighbourhood matrix that [INLA](#) needs, and fit a spatiotemporal model with seasonal, long-term and BYM2 spatial random effects;
 3. Allow the effects of temperature and rainfall to vary across space, and map the exceedance probabilities; and
 4. Fit a distributed lag non-linear model in [INLA](#), so that the effect of temperature can be both non-linear and delayed.

Questions?